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	KET	BRK
	$ \omega\rangle = \sum_{N=1}^d \omega_N \phi_N\rangle$	$\langle\omega = \sum_{N=1}^d \omega_N^* \langle\phi_N $
BASIS	$\{ \phi_N\rangle\}$	$\{\langle\phi_N \}$
COMPONENTS	$\omega = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_d \end{pmatrix}$	$\omega^\dagger = (\omega_1^* \dots \omega_d^*)$

OPERATOR VECTOR $\rightarrow$ VECTOR

AN OPERATOR G ON THE HILBERT SPACE $\mathcal{H}$ IS A FUNCTION THAT MAPS VECTORS TO VECTORS IN A LINEAR WAY

$$G\left(\sum_{N=1}^d \omega_N |\phi_N\rangle\right)$$

SOME IMPORTANT OPERATORS

• IDENTITY OPERATOR $1|\psi\rangle = |\psi\rangle$

• COMMUTATOR BETWEEN TWO OPERATORS A AND B

$$[A, B] = AB - BA$$

$$[A, B]|\psi\rangle = (AB - BA)|\psi\rangle = A(B|\psi\rangle) - B(A|\psi\rangle)$$

NOTE THAT $[A, B] = -[B, A]$ AND $[A, A] = 0$

• OUTER PRODUCT

$$|\omega\rangle\langle\beta| \quad |\alpha\rangle\langle\beta| (|\psi\rangle) = |\alpha\rangle\langle\beta|\psi\rangle$$

A AND B
 $\downarrow$ $\downarrow$
 S_x S_z

start
 $\downarrow$
 $|\chi_+\rangle = \frac{1}{\sqrt{2}}(|z_+\rangle + \frac{1}{\sqrt{2}}|z_-\rangle)$

$$[S_y, S_z] = S_y S_z - S_z S_y$$

$$[A, B] = -[B, A]$$

$$S_x S_z |\chi_+\rangle - S_z S_x |\chi_+\rangle = ?$$

= 0
THEY
COMMUTE

NO
THEY NOT
COMMUTE

CONSIDER THE QUBIT HILBERT SPACE $\mathcal{H}$ WITH STANDARD BASIS VECTORS $|0\rangle$ AND $|1\rangle$. LET $A = |0\rangle\langle 1|$ AND

$B = |1\rangle\langle 0|$ SHOW THAT $AB \neq BA$

$$A = |0\rangle\langle 1|$$

$$B = |1\rangle\langle 0|$$

$$[A, B] \neq 0$$

$$[A, B] = \underbrace{|0\rangle\langle 1|}_{A} \underbrace{|1\rangle\langle 0|}_{B} - \underbrace{|1\rangle\langle 0|}_{B} \underbrace{|0\rangle\langle 1|}_{A}$$

$$= |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$= |0\rangle\langle 0| - |1\rangle\langle 1| \neq 0$$

INNER PRODUCT $\rightarrow 1$

PROJECT ANY STATE ON THIS
PROJECTOR ON $|\alpha\rangle \quad \Pi_\alpha = |\alpha\rangle\langle\alpha|$

CONSIDER $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$, CALCULATE $\Pi_0|\psi\rangle$

$$\begin{aligned} \Pi_0|\psi\rangle &= |0\rangle\langle 0|\psi\rangle = |0\rangle\langle 0|(c_0|0\rangle + c_1|1\rangle) = \\ &= c_0|0\rangle\langle 0|0\rangle + c_1|0\rangle\langle 0|1\rangle = c_0|0\rangle \end{aligned}$$

COMPLETENESS RELATION

$\{|N\rangle\}$ ORTHONORMAL BASIS FOR $\mathcal{H}$ (i.e. $\langle M|N\rangle = \delta_{MN}$)

$$\sum_N \Pi_N = \mathbb{1} \quad \text{IDENTITY} = \sum_N |N\rangle\langle N|$$

WE WANT TO PROVE $\sum_N |N\rangle\langle N| = \mathbb{1} \quad \{|N\rangle\}$

$$|\psi\rangle = \sum_N \omega_N |N\rangle$$

$$\begin{aligned} \left(\sum_N |N\rangle\langle N|\right)|\psi\rangle &= \left(\sum_N |N\rangle\langle N|\right)\left(\sum_N \omega_N |N\rangle\right) = \sum_N \sum_M \omega_N |N\rangle\langle N|M\rangle = \sum_N \sum_M \omega_N \delta_{NM} \\ &= \sum_N \omega_N |N\rangle = |\psi\rangle \end{aligned}$$

PAULI OPERATOR

FOR THE QUBIT HILBERT SPACE $\mathcal{Q}$ WE CAN INTRODUCE SOME USEFUL OPERATORS

IDENTITY OPERATOR: $I = |0\rangle\langle 0| + |1\rangle\langle 1|$

PAULI OPERATOR:

$$\begin{aligned} X &= |0\rangle\langle 1| + |1\rangle\langle 0| \\ Y &= -i|0\rangle\langle 1| + i|1\rangle\langle 0| \\ Z &= |0\rangle\langle 0| - |1\rangle\langle 1| \end{aligned}$$

THESE OPERATORS DEPEND ON THE CHOSEN BASIS

CALCULATE $X|0\rangle$, $X|1\rangle$, $Y|0\rangle$, $Y|1\rangle$, $Z|0\rangle$ AND $Z|1\rangle$

$$X|0\rangle = (|0\rangle\langle 1| + |1\rangle\langle 0|)|0\rangle = |1\rangle$$

$$X|1\rangle = (|0\rangle\langle 1| + |1\rangle\langle 0|)|1\rangle = |0\rangle$$

$$Y|0\rangle = (-i|0\rangle\langle 1| + i|1\rangle\langle 0|)|0\rangle = i|1\rangle$$

$$Y|1\rangle = -i|0\rangle$$

$$Z|0\rangle = (|0\rangle\langle 0| - |1\rangle\langle 1|)|0\rangle = |0\rangle$$

$$Z|1\rangle = (|0\rangle\langle 0| - |1\rangle\langle 1|)|1\rangle = -|1\rangle$$

$$\begin{aligned} |1\rangle &\rightarrow |Z-\rangle \\ |0\rangle &\rightarrow |Z+\rangle \end{aligned} \quad S_z|Z-\rangle = -\hbar/2|Z-\rangle$$

$$S_z|Z+\rangle = +\hbar/2|Z+\rangle$$

SHOW THAT $X^2 = Y^2 = Z^2 = I$

$$X^2 = (|0\rangle\langle 1| + |1\rangle\langle 0|)(|0\rangle\langle 1| + |1\rangle\langle 0|) = \cancel{|0\rangle\langle 1|0\rangle\langle 1|} + \overset{1}{|0\rangle\langle 1|1\rangle\langle 0|} + \overset{1}{|1\rangle\langle 0|0\rangle\langle 1|} + \cancel{|1\rangle\langle 0|1\rangle\langle 0|} = |0\rangle\langle 0| + |1\rangle\langle 1| = I$$

$$\begin{aligned} Y^2 &= (i|0\rangle\langle 1| - i|1\rangle\langle 0|)(-i|0\rangle\langle 1| + i|1\rangle\langle 0|) = -i^2|0\rangle\langle 1|1\rangle\langle 0| - i^2|1\rangle\langle 0|0\rangle\langle 1| = \\ &= |0\rangle\langle 0| + |1\rangle\langle 1| = I \end{aligned}$$

SHOW THAT $XY = iZ$ (AND CYCLIC PERMUTATIONS)

$$(|0\rangle\langle 1| + |1\rangle\langle 0|)(-i|0\rangle\langle 1| + i|1\rangle\langle 0|) = i|0\rangle\langle 0| - i|1\rangle\langle 1| = i(|0\rangle\langle 0| - |1\rangle\langle 1|) = iZ$$

$$\hat{Z}X = iY \quad YZ = iX$$

OPERATOR AND MATRICES

$$\{|N\rangle\} \longrightarrow G \left(\sum_{n=1}^d a_n |N\rangle \right) = \sum_{n=1}^d a_n G(|N\rangle)$$

GIVEN THE BASIS WE CAN SPECIFY THE OPERATOR USING THE $d \times d$ MATRIX

$$G = (G(|1\rangle) \quad G(|2\rangle) \quad \dots \quad G(|N\rangle))$$

$$G_{mn} = \langle m | G | n \rangle \Leftrightarrow G_{mn} = \langle m | G | n \rangle \quad G(|1\rangle) = G|1\rangle$$

LIKEWISE THE VECTOR CASE, THE MATRIX G IS CALLED THE REPRESENTATION OF THE OPERATOR G IN THE BASIS $\{|N\rangle\}$

FOR THE PAULI OPERATIONS AND THE $|0\rangle$, $|1\rangle$ BASIS WE HAVE

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

PAULI MATRICES

$$Y = -i|0\rangle\langle 1| + i|1\rangle\langle 0| \quad \text{OPERATOR}$$

$$\{|0\rangle, |1\rangle\} \text{ BASIS}$$

$$Y_{MN} = \langle M|Y|N\rangle$$

$$Y_{00} = \langle 0|Y|0\rangle = \langle 0|(-i|0\rangle\langle 1| + i|1\rangle\langle 0|)|0\rangle = \langle 0|i|1\rangle = i\langle 0|1\rangle = 0$$

$$Y_{01} = \langle 0|Y|1\rangle = \langle 0|(-i|0\rangle\langle 1| + i|1\rangle\langle 0|)|1\rangle = \langle 0|-i|0\rangle = -i\langle 0|0\rangle = -i$$

$$Y_{10} = \langle 1|Y|0\rangle = \langle 1|(-i|0\rangle\langle 1| + i|1\rangle\langle 0|)|0\rangle = \langle 1|-i|1\rangle = -i\langle 1|1\rangle = -i$$

$$Y_{11} = 0$$

$$\begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} \downarrow \\ 0 \\ \downarrow \\ 1 \end{matrix} & \begin{bmatrix} Y_{00} & Y_{01} \\ Y_{10} & Y_{11} \end{bmatrix} \end{matrix}$$

OPERATOR AND MATRICES

GIVEN A CERTAIN BASIS $\{|N\rangle\}$ THE RESULT $|\beta\rangle$ OF THE ACTION OF THE OPERATOR G ON THE VECTOR $|\alpha\rangle$ IS REPRESENTED

BY THE PRODUCT OF THE REPRESENTATION OF G AND $|\alpha\rangle$

$$|\beta\rangle = G|\alpha\rangle \rightarrow \begin{pmatrix} b_1 \\ \vdots \\ b_d \end{pmatrix} = \begin{pmatrix} G_{11} & \dots & G_{1d} \\ \vdots & & \vdots \\ G_{d1} & \dots & G_{dd} \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_d \end{pmatrix} \quad b_N = \langle N|\beta\rangle = \sum_N \langle N|G|N\rangle \langle N|\alpha\rangle$$

$$G = \sum_{MN} |M\rangle \langle M|G|N\rangle \langle N| = \sum_{MN} G_{MN} |M\rangle \langle N| \quad G_{MN} = \langle M|G|N\rangle$$

$$b_N = \langle N|\beta\rangle = \sum_N \langle N|G|N\rangle \langle N|\alpha\rangle = \sum_N \underbrace{G_{NN}}_{\text{MATRIX ELEMENT}} a_N$$

$$\langle N|\alpha\rangle = a_N \quad |\alpha\rangle = \sum_N a_N |N\rangle$$

$$\langle N|G|N\rangle = G_{NN}$$

$$\begin{pmatrix} (AB)_{11} & \dots & (AB)_{1d} \\ \vdots & & \vdots \\ (AB)_{d1} & \dots & (AB)_{dd} \end{pmatrix} = \begin{pmatrix} A_{11} & \dots & A_{1d} \\ \vdots & & \vdots \\ A_{d1} & \dots & A_{dd} \end{pmatrix} \begin{pmatrix} B_{11} & \dots & B_{1d} \\ \vdots & & \vdots \\ B_{d1} & \dots & B_{dd} \end{pmatrix}$$

WE CAN ALSO USE THE MORE COMPACT FORM

$$(AB)_{NN} = \langle N|AB|N\rangle = \sum_K \langle N|A|K\rangle \langle K|B|N\rangle = \sum_K A_{NK} B_{KN}$$

$$\text{IMPORTANT: IN GENERAL ONE HAS THAT } \sum_K A_{NK} B_{KN} \neq \sum_N B_{NS} A_{SN} \quad AB \neq BA$$

THE TRACE IS LINEAR FUNCTIONAL ON OPERATOR $TR: \mathcal{O} \rightarrow \mathbb{C}$

$$TRA = \sum_N \langle N | A | N \rangle = \sum_N A_{NN}$$

SOME PROPERTIES

- THE TRACE IS INDEPENDENT OF THE BASIS OF THE REPRESENTATION
- CYCLIC PROPERTY $TRA \dots B_C = TR CA \dots B$
- THE TRACE IS LINEAR AND THE TRACE ON ANY COMMUTATOR IS ZERO

IN A QUBIT HILBERT SPACE, LET $A = |0\rangle\langle 0|$, $B = |0\rangle\langle 1|$ AND $C = |1\rangle\langle 0|$

SHOW THAT $TR(ABC) \neq TR(ACB)$

$$A = |0\rangle\langle 0| \quad B = |0\rangle\langle 1| \quad C = |1\rangle\langle 0|$$

- OPTION 1: FIND THE REPRESENTATION OF A, B, C
THEN FIND ABC AND TRACE ABC

- OPTION 2: FIND ABC AND FIND THE REPRESENTATION
OF ABC AND TRACE

- OPTION 3: FIND $ABC \rightarrow$ APPLY DEFINITION $TR(A) = \sum_N \langle N | A | N \rangle$

$$\rightarrow ABC = (|0\rangle\langle 0| |0\rangle\langle 1| |1\rangle\langle 0|) = |0\rangle\langle 0|$$

IN QUANTUM MECHANICS, AN OBSERVABLE IS A PROPERTY OF THE SYSTEM

